Title: Simplifying product selection

Start date: //_____

Finish date: //_____

Responsible: Yuliia Stepanova

Description

Test goal: increase `add_to_cart` by 5%
Hypothesis: We believe that if we make changes to the product recommendations to the buyer based on his previous purchases and offer him only one product instead of three, we will simplify his choice and increase the percentage of adding to the cart.

Options

Option A (Control): unchanged.
Option B (Experimental): with a recommendation of only one product based on the user's previous orders.

Data sampling

Target audience: all site visitors.
Traffic split: 50% for option A, 50% for option B

Metrics

Primary metric: `add_to_cart` / `session.`
Additional metrics: we do not use.

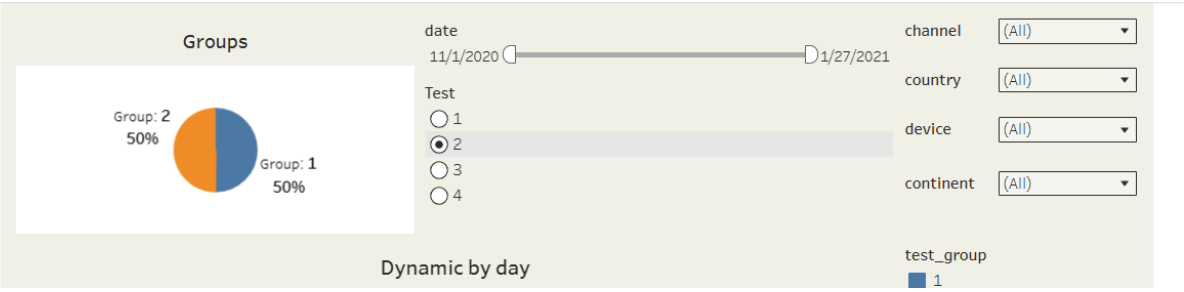
Results

Test results for all metrics:

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	2,344	2,409	2.77%
add_shipping_info	3,480	3,510	0.86%
add_to_cart	2,811	3,061	8.89%
begin_checkout	4,262	4,313	1.20%
first_visit	34,511	34,171	-0.99%
new account	4,165	4,184	0.46%
page_view	220,275	212,320	-3.61%
scroll	80,713	81,370	0.81%
select_promotion	1,477	1,406	-4.81%
session	50,637	50,244	-0.78%
session with orders	5,102	5,003	-1.94%
session_start	51,219	50,808	-0.80%
user_engagement	198,266	189,931	-4.20%
view_item	72,717	68,700	-5.52%
view_promotion	32,367	31,680	-2.12%

Analyze

Checking the 50% to 50% distribution:



Change in key metrics: we check the changes in `add_to_cart` / `session` for statistical significance:

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes		# trials	Confidence interval	[clear] [link]
Sample 1:	2811	/	50637		*5.4% – 5.8%*
Sample 2:	3061	/	50244		*5.9% – 6.3%*

Verdict:

Sample 2 is more successful

($p < 0.001$)

Confidence level: 95%

If the experiment is repeated many times, the confidence level is the percent of the time each sample's success rate will fall within the reported confidence interval.

We see a statistically significant increase in the indicator by 10%:

	test_group		
event_name	1	2	
add_to_cart	2811	3061	
session	50637	50244	
	5,55%	6,09%	10%

Changes by segments: Let's review separately by devices, channels, and continents.

A statistically insignificant drop in the indicator was found for the *Direct* and *Undefined* channels.

Across continents, there is a significant drop in metrics in Asia:

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	572	630	10.14%
add_shipping_info	833	871	4.56%
add_to_cart	752	619	-17.69%
begin_checkout	1,008	1,102	9.33%
first_visit	8,137	8,097	-0.49%
new account	985	1,025	4.06%
page_view	51,672	49,739	-3.74%
scroll	19,060	19,289	1.20%
select_promotion	374	314	-16.04%
session	11,983	11,983	0.00%
session with orders	1,187	1,211	2.02%
session_start	12,121	12,093	-0.23%
user_engagement	46,284	44,570	-3.70%
view_item	17,113	16,009	-6.45%
view_promotion	7,629	7,442	-2.45%

Evan's Awesome A/B Tools (home):

Sample Size Calculator | **Chi-Squared Test** | Sequential Sampling | 2 Sample T-Test | Survival Times | Count Data

Question: Does the rate of success differ across two groups?

	# successes		# trials	Confidence interval	[clear] [link]
Sample 1:	752	/	11983		*5.9% – 6.7%*
Sample 2:	619	/	11983		*4.8% – 5.6%*

Verdict:

Sample 1 is more successful

($p < 0.001$)

Confidence level: 95%

If the experiment is repeated many times, the confidence level is the percent of the time each sample's success rate will fall within

Conclusions and recommendations

Next steps: Based on the test results, it is recommended to make changes to the recommendations. It is proposed to take into account the statistically significant negative results for users from Asia. It is recommended to leave them with a choice of recommendations from three options. In this case, we can expect a much larger increase in the metric.

	value		value (%)
	test_group		test_group
event_name	1	2	2
add_payment_info	1,772	1,779	0.40%
add_shipping_info	2,647	2,639	-0.30%
add_to_cart	2,059	2,442	18.60%
begin_checkout	3,254	3,211	-1.32%
first_visit	26,374	26,074	-1.14%
new account	3,180	3,159	-0.66%
page_view	168,603	162,581	-3.57%
scroll	61,653	62,081	0.69%
select_promotion	1,103	1,092	-1.00%
session	38,654	38,261	-1.02%
session with orders	3,915	3,792	-3.14%
session_start	39,098	38,715	-0.98%
user_engagement	151,982	145,361	-4.36%
view_item	55,604	52,691	-5.24%
view_promotion	24,738	24,238	-2.02%